

ACTIVITY SLOT A

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UNIT IV – CLASS ACTIVITY CODES

1. Study Hours vs Marks — R Scatterplot

Task: Create dataset, draw scatterplot, add labels/title, interpret positive correlation.

```
# Unit IV - Question 1
```

```
# Study Hours vs Marks
```

```
study_hours <- c(2, 3, 4, 5, 6, 7, 8, 9)
```

```
marks <- c(45, 52, 58, 63, 69, 74, 81, 88)
```

```
data <- data.frame(study_hours, marks)
```

```
print(data)
```

```
# Scatterplot
```

```
plot(
```

```
  data$study_hours,
```

```
  data$marks,
```

```
  main = "Study Hours vs Marks",
```

```
  xlab = "Study Hours",
```

```
  ylab = "Marks",
```

```
  pch = 19
```

```
)
```

```
# Add regression line
```

```
abline(lm(marks ~ study_hours, data = data))
```

```
# Correlation
```

```
correlation <- cor(study_hours, marks)
```

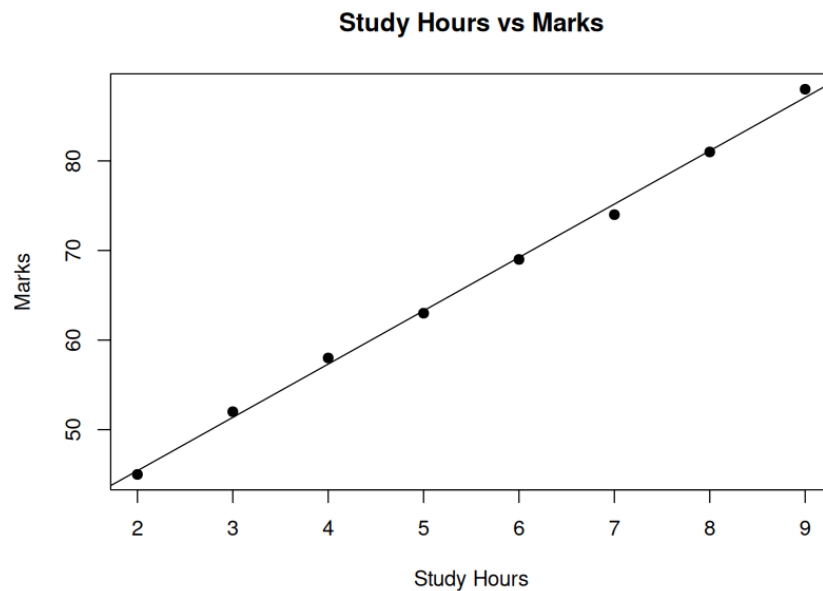
```
print(paste("Correlation =", correlation))
```

Interpretation:

The scatterplot shows an upward trend. Therefore, a strong positive correlation exists between study hours and marks. As study hours increase, marks also tend to increase.

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2. Age vs Systolic Blood Pressure — Python Matplotlib

Unit IV - Question 2

Age vs Systolic Blood Pressure

```
import matplotlib.pyplot as plt
```

```
age = [20, 25, 30, 35, 40, 45, 50, 55, 60, 65]
```

```
blood_pressure = [110, 112, 115, 118, 122, 125, 128, 132, 135, 140]
```

```
plt.scatter(age, blood_pressure, color='red')
```

```
plt.xlabel("Age")
```

```
plt.ylabel("Systolic Blood Pressure")
```

```
plt.title("Age vs Systolic Blood Pressure")
```

```
plt.grid(True)
```

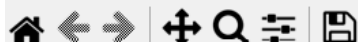
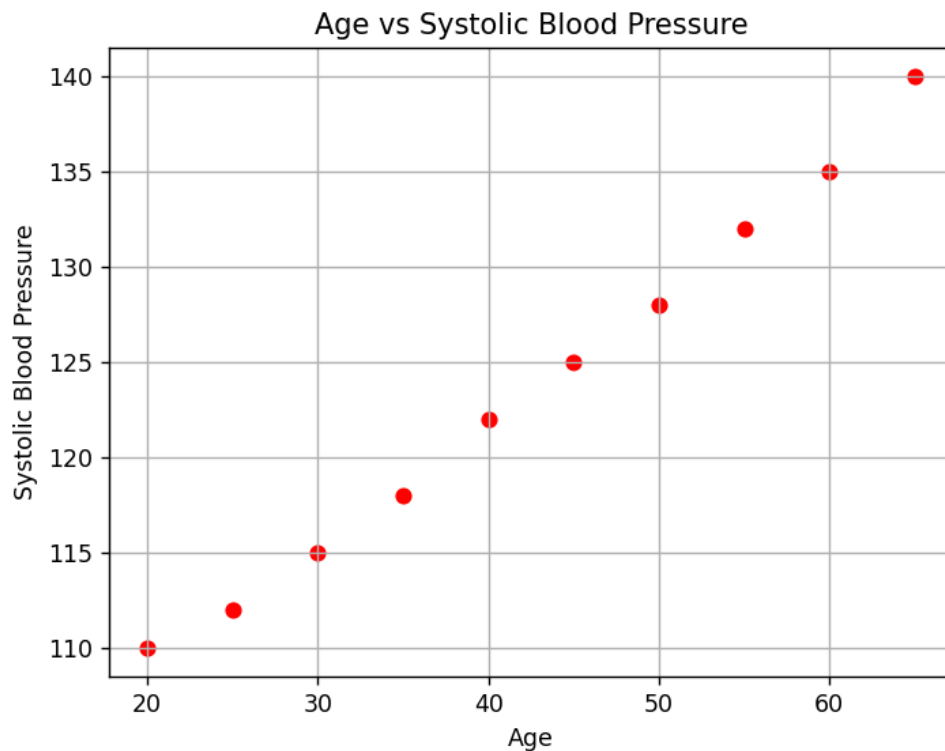
```
plt.show()
```

Comment:

The scatterplot shows a positive trend. Systolic blood pressure generally increases as age increases.

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(x, y) = (25.89, 141.07)

3. Advertising Expenditure vs Monthly Sales — Python

Unit IV - Question 3

Advertising Expenditure vs Monthly Sales

```
import matplotlib.pyplot as plt
```

```
advertising = [10, 15, 20, 25, 30, 35, 40, 45, 50]
```

```
sales = [100, 120, 135, 150, 175, 190, 210, 230, 250]
```

```
plt.scatter(advertising, sales)
```

```
plt.xlabel("Advertising Expenditure")
```

```
plt.ylabel("Monthly Sales")
```

```
plt.title("Advertising Expenditure vs Monthly Sales")
```

```
plt.grid(True)
```

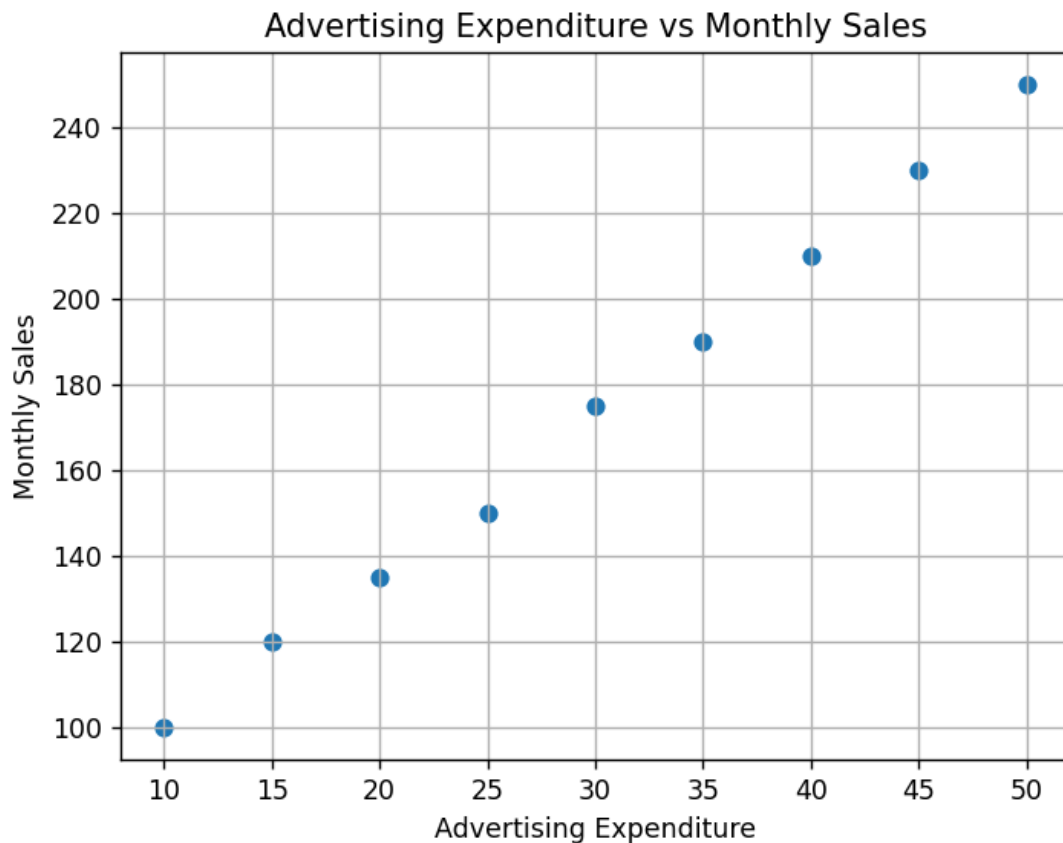
```
plt.show()
```

Interpretation:

The scatterplot indicates a positive relationship. Monthly sales generally increase as advertising expenditure increases.

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4. Financial Dataset — Correlation Matrix and Correlogram in R

Variables:

Income

Savings

Loan Amount

Credit Score

Unit IV - Question 4

Financial Data Correlation Matrix

```
income <- c(25000, 30000, 35000, 40000, 45000, 50000, 55000, 60000, 65000, 70000)
```

```
savings <- c(5000, 7000, 9000, 12000, 14000, 16000, 19000, 22000, 25000, 28000)
```

```
loan_amount <- c(15000, 18000, 22000, 25000, 28000, 30000, 35000, 38000, 42000, 45000)
```

```
credit_score <- c(620, 630, 640, 650, 665, 675, 685, 700, 715, 730)
```

```
financial_data <- data.frame(
```

```
  Income = income,
```

```
  Savings = savings,
```

```
  Loan_Amount = loan_amount,
```

```
  Credit_Score = credit_score
```

```
)
```

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```
print(financial_data)
```

```
# Correlation matrix
```

```
cor_matrix <- cor(financial_data)
```

```
print(cor_matrix)
```

```
# Correlogram
```

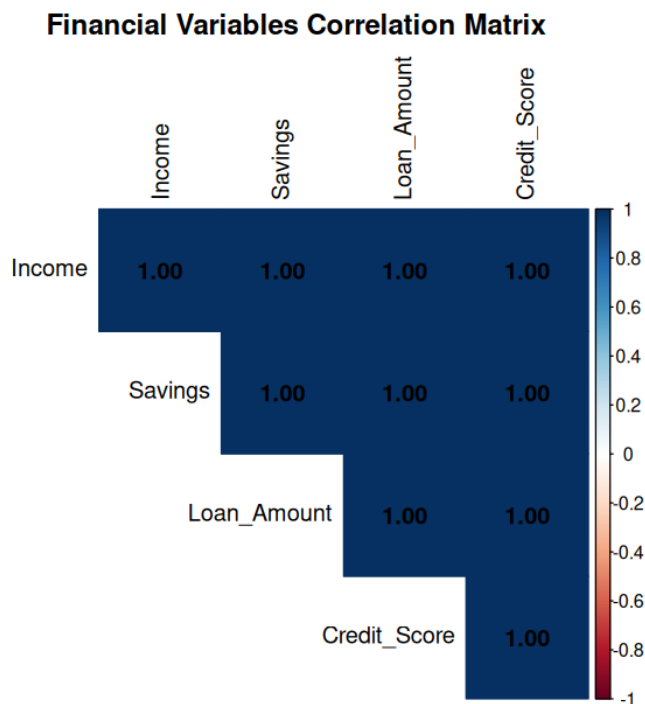
```
install.packages("corrplot")
```

```
library(corrplot)
```

```
corrplot(  
  cor_matrix,  
  method = "color",  
  type = "upper",  
  addCoef.col = "black",  
  tl.col = "black",  
  title = "Financial Variables Correlation Matrix",  
  mar = c(0, 0, 2, 0)  
)
```

Interpretation:

The correlation matrix identifies the relationships among income, savings, loan amount and credit score. The variables with correlation values closest to +1 exhibit the strongest positive correlation.



5. Financial Dataset — Python Correlogram

This is the duplicate financial question, so you can use this Python version.

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Unit IV - Question 5

Financial Data Correlation Matrix

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

data = {
    "Income": [25000, 30000, 35000, 40000, 45000, 50000, 55000, 60000, 65000, 70000],
    "Savings": [5000, 7000, 9000, 12000, 14000, 16000, 19000, 22000, 25000, 28000],
    "Loan Amount": [15000, 18000, 22000, 25000, 28000, 30000, 35000, 38000, 42000, 45000],
    "Credit Score": [620, 630, 640, 650, 665, 675, 685, 700, 715, 730]
}

df = pd.DataFrame(data)

correlation = df.corr()

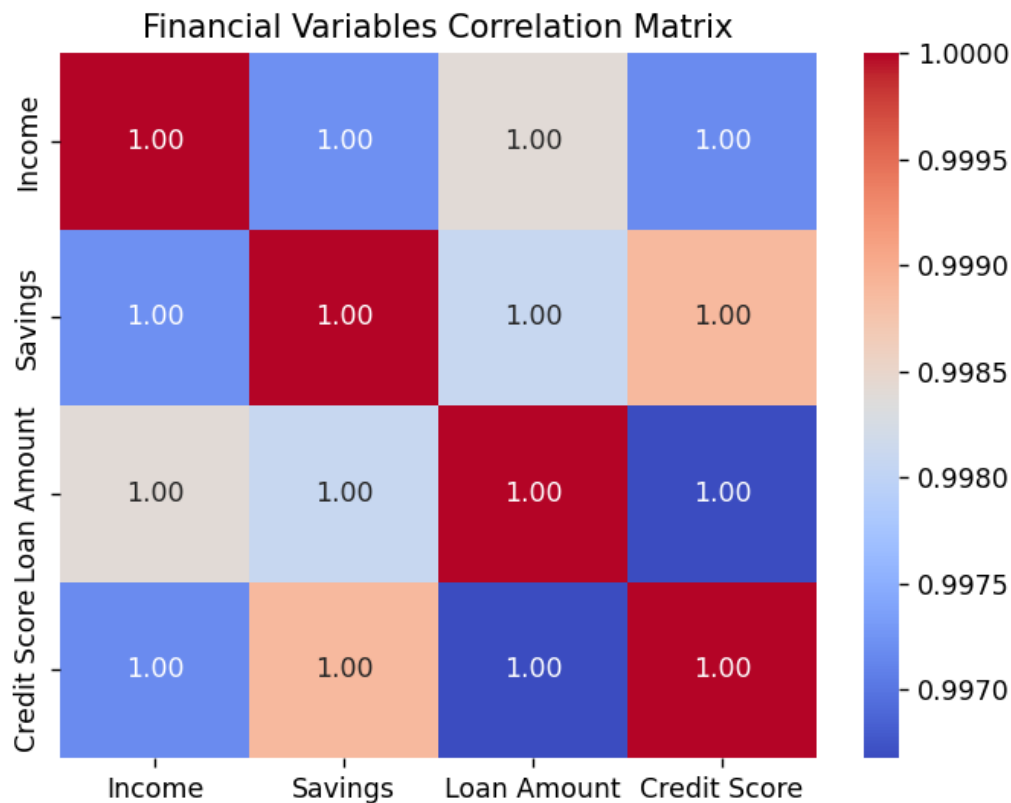
print(correlation)

sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Financial Variables Correlation Matrix")
plt.show()
```

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6. Iris Dataset — Correlation Matrix and Heatmap

Unit IV - Question 6

Iris Correlation Matrix

```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
iris = sns.load_dataset("iris")
```

```
# Select numerical columns
numeric_data = iris.select_dtypes(include="number")
```

```
# Correlation matrix
correlation = numeric_data.corr()
```

```
print(correlation)
```

```
# Heatmap
sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
```

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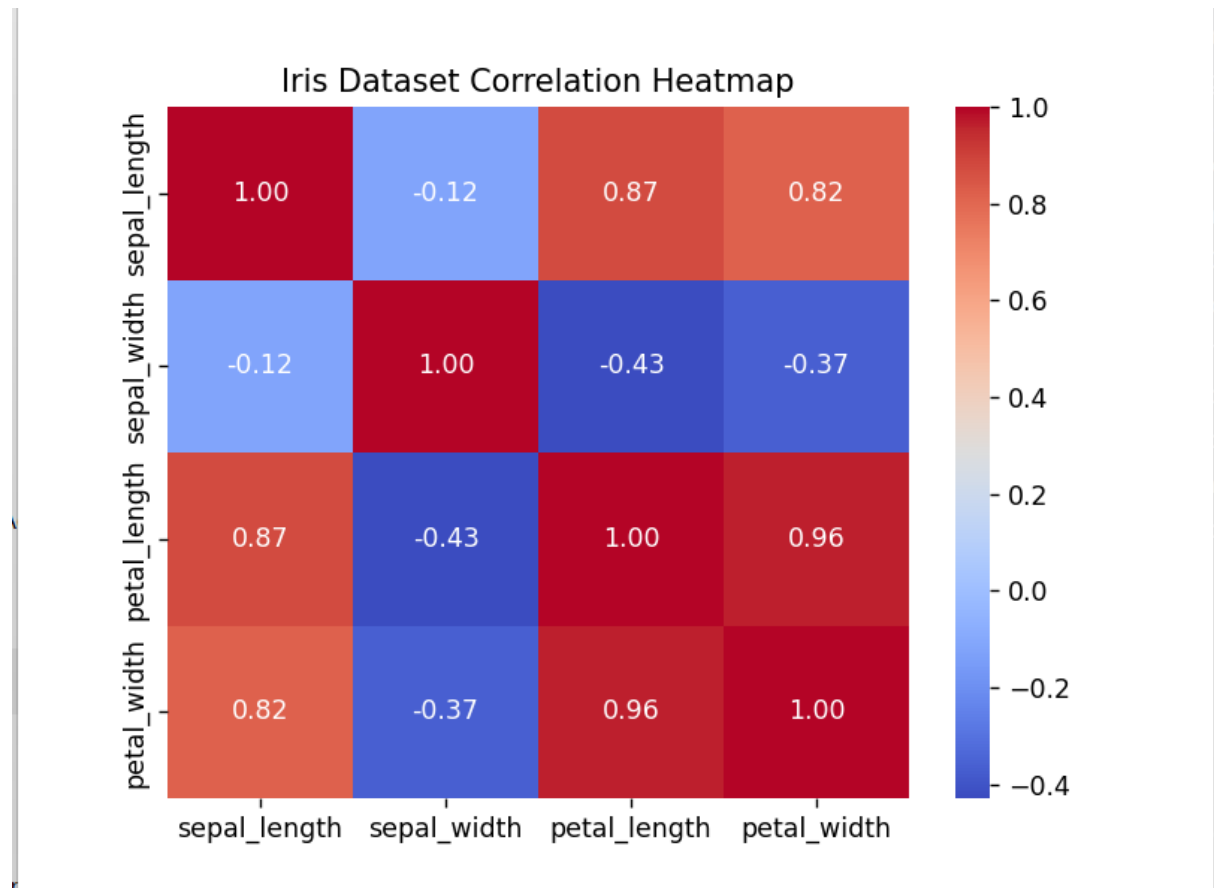
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)

```
plt.title("Iris Dataset Correlation Heatmap")  
plt.show()
```

Negative correlation:

In the Iris dataset, Sepal Length and Sepal Width show a negative correlation.



7. Healthcare Dataset — Correlogram and Multicollinearity

Unit IV - Question 7

Healthcare Correlation Analysis

```
import pandas as pd  
import seaborn as sns  
import matplotlib.pyplot as plt
```

```
data = {  
    "BMI": [22, 24, 25, 27, 28, 30, 31, 33, 35, 36],  
    "Cholesterol": [170, 175, 180, 190, 195, 205, 210, 220, 230, 240],  
    "Blood Pressure": [110, 112, 115, 118, 120, 125, 128, 132, 135, 140],  
    "Glucose": [85, 88, 90, 95, 100, 105, 110, 115, 120, 125],  
    "Heart Rate": [68, 70, 72, 74, 75, 77, 79, 80, 82, 84]  
}
```


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```
df = pd.DataFrame(data)
```

```
correlation = df.corr()
```

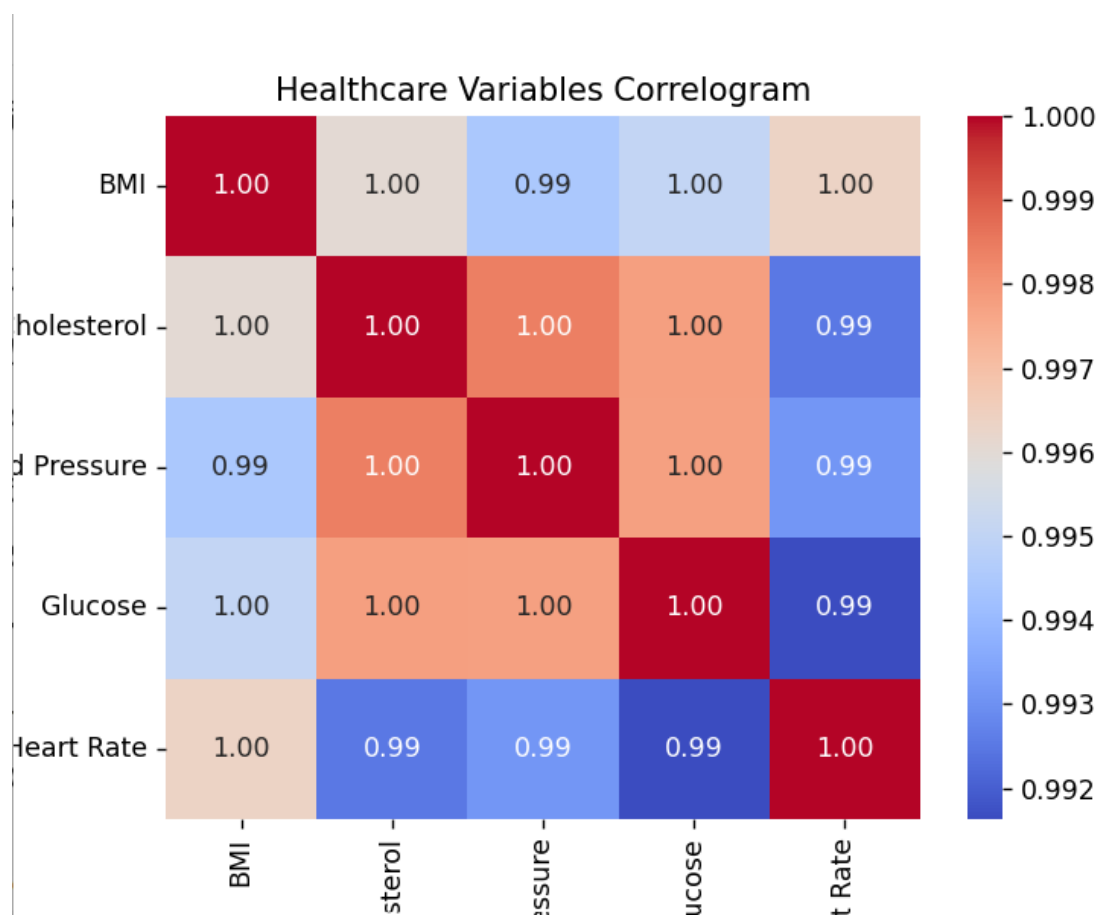
```
print(correlation)
```

```
sns.heatmap(  
    correlation,  
    annot=True,  
    cmap="coolwarm",  
    fmt=".2f"  
)
```

```
plt.title("Healthcare Variables Correlogram")  
plt.show()
```

Interpretation:

Strong correlations between independent variables may indicate multicollinearity. Correlation values close to +1 or -1 indicate strong relationships.



8. MRI Dataset — PCA to Two Principal Components

Unit IV - Question 8

PCA on 25 MRI Features

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```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler

np.random.seed(42)

# Simulated MRI dataset: 100 samples, 25 features
data = np.random.rand(100, 25)

# Standardization
scaled_data = StandardScaler().fit_transform(data)

# PCA
pca = PCA(n_components=2)
pca_data = pca.fit_transform(scaled_data)

print("Explained Variance Ratio:")
print(pca.explained_variance_ratio_)

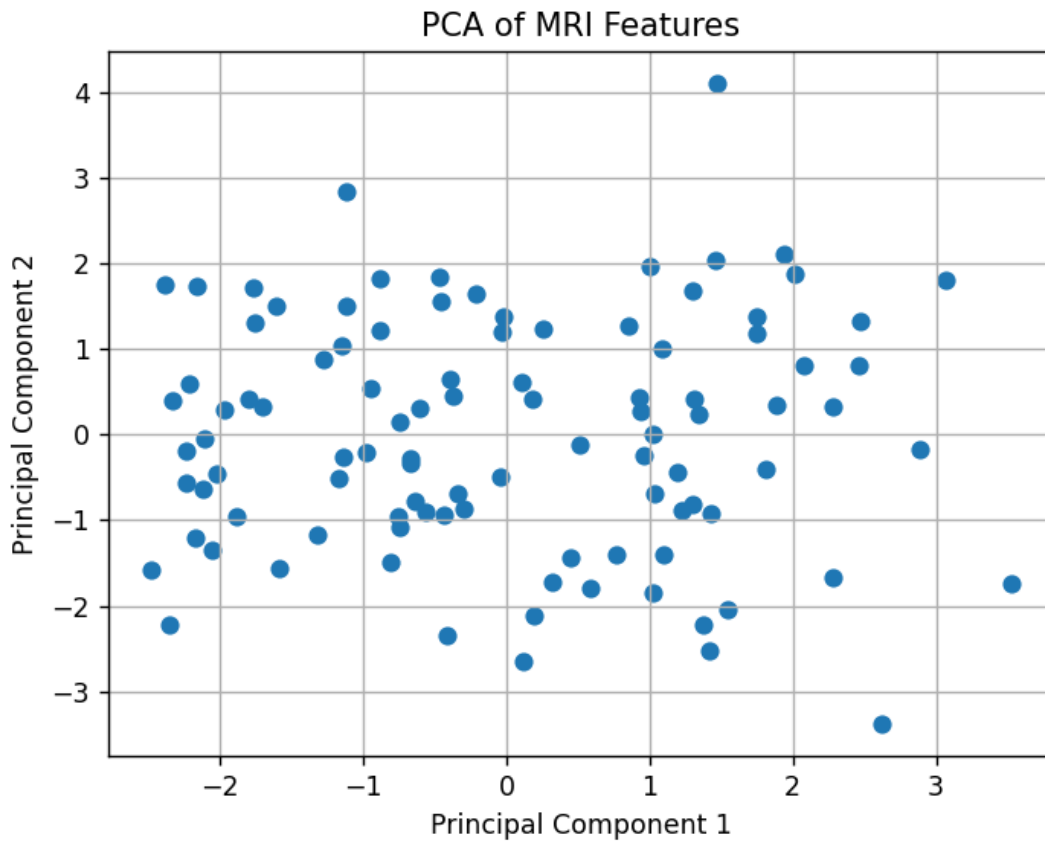
# Plot
plt.scatter(pca_data[:, 0], pca_data[:, 1])

plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("PCA of MRI Features")
plt.grid(True)

plt.show()
```

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9. Iris Dataset — PCA Summary and Biplot

Unit IV - Question 9

PCA on Iris Dataset

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
```

```
iris = load_iris()
```

```
X = iris.data
```

```
y = iris.target
```

```
# Standardize
```

```
X_scaled = StandardScaler().fit_transform(X)
```

```
# PCA
```

```
pca = PCA(n_components=2)
```

```
X_pca = pca.fit_transform(X_scaled)
```

```
# Summary
```

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```
print("Explained Variance Ratio:")
print(pca.explained_variance_ratio_)
```

```
print("\nTotal Variance Explained:")
print(sum(pca.explained_variance_ratio_))
```

```
# Biplot
plt.scatter(
    X_pca[:, 0],
    X_pca[:, 1],
    c=y
)
```

```
# Feature vectors
features = iris.feature_names
```

```
for i, feature in enumerate(features):
    plt.arrow(
        0, 0,
        pca.components_[0, i] * 2,
        pca.components_[1, i] * 2,
        head_width=0.05
    )
```

```
plt.text(
    pca.components_[0, i] * 2.2,
    pca.components_[1, i] * 2.2,
    feature
)
```

```
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("PCA Biplot - Iris Dataset")
plt.grid(True)
```

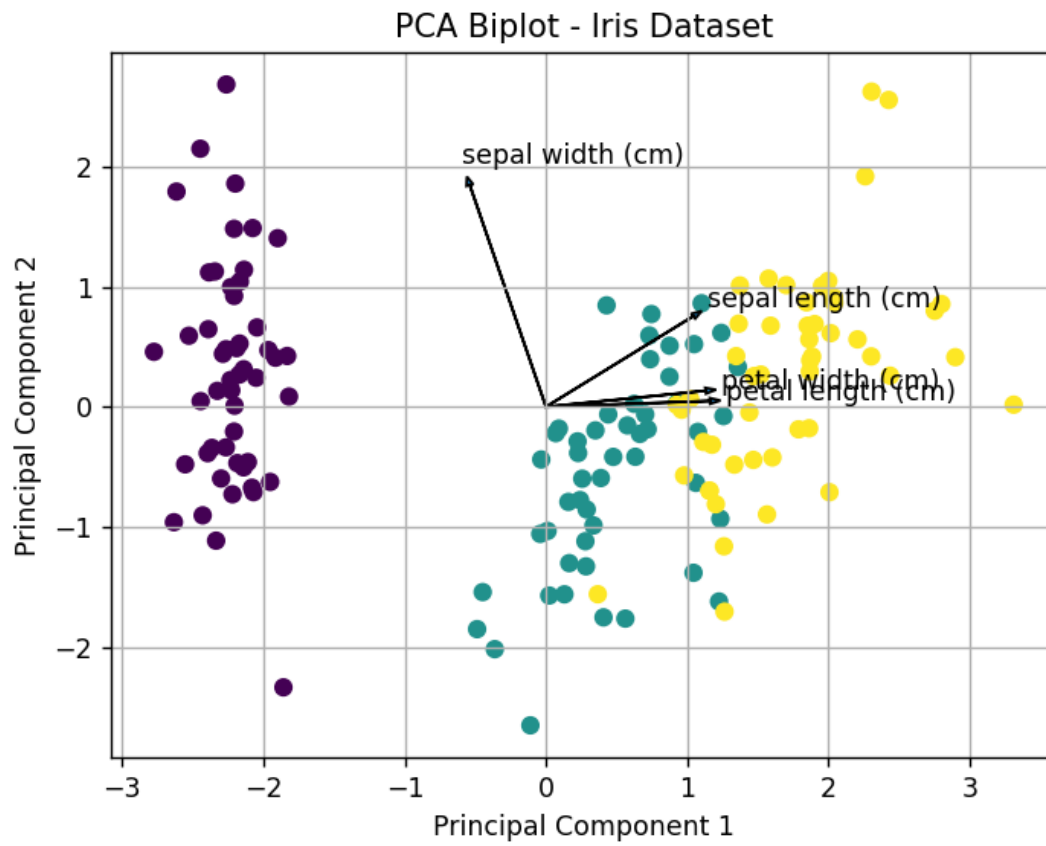
```
plt.show()
```

Explanation:

Principal Component 1 (PC1) explains the highest variance because it has the largest explained variance ratio.

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10. Automobile Dataset — PCA

Unit IV - Question 10

PCA on 15 Automobile Specifications

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
```

```
np.random.seed(42)
```

```
# 100 cars and 15 specifications
data = np.random.rand(100, 15)
```

```
# Standardization
scaled_data = StandardScaler().fit_transform(data)
```

```
# PCA
pca = PCA(n_components=2)
pca_data = pca.fit_transform(scaled_data)
```

```
print("Explained Variance:")
print(pca.explained_variance_ratio_)
```

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Visualization

```
plt.scatter(pca_data[:, 0], pca_data[:, 1])
```

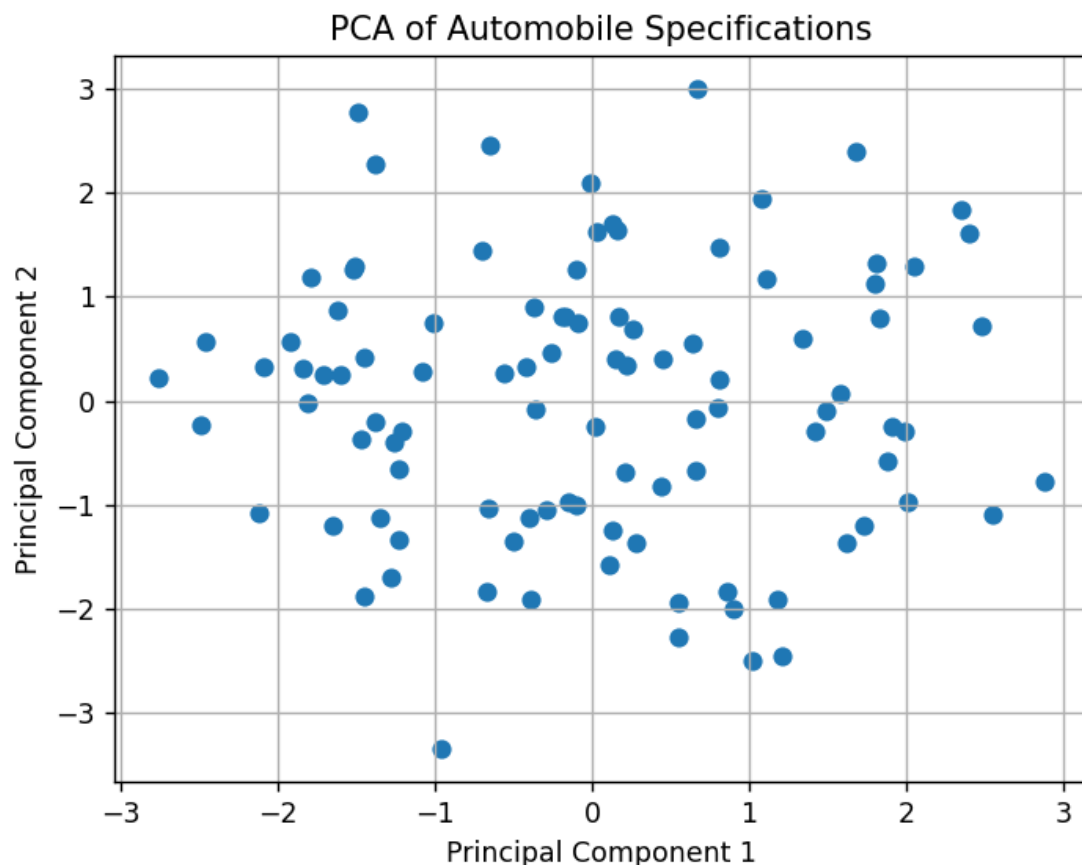
```
plt.xlabel("Principal Component 1")
```

```
plt.ylabel("Principal Component 2")
```

```
plt.title("PCA of Automobile Specifications")
```

```
plt.grid(True)
```

```
plt.show()
```



11. Facial Recognition — t-SNE

Unit IV - Question 11

t-SNE on 128-Dimensional Facial Embeddings

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
from sklearn.manifold import TSNE
```

```
from sklearn.preprocessing import StandardScaler
```

```
np.random.seed(42)
```

150 facial samples with 128 features

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```
data = np.random.rand(150, 128)
```

```
# Standardization
```

```
data_scaled = StandardScaler().fit_transform(data)
```

```
# t-SNE
```

```
tsne = TSNE(  
    n_components=2,  
    random_state=42,  
    perplexity=30  
)
```

```
data_tsne = tsne.fit_transform(data_scaled)
```

```
# Plot
```

```
plt.scatter(  
    data_tsne[:, 0],  
    data_tsne[:, 1]  
)
```

```
plt.xlabel("t-SNE Dimension 1")
```

```
plt.ylabel("t-SNE Dimension 2")
```

```
plt.title("t-SNE Visualization of Facial Embeddings")
```

```
plt.grid(True)
```

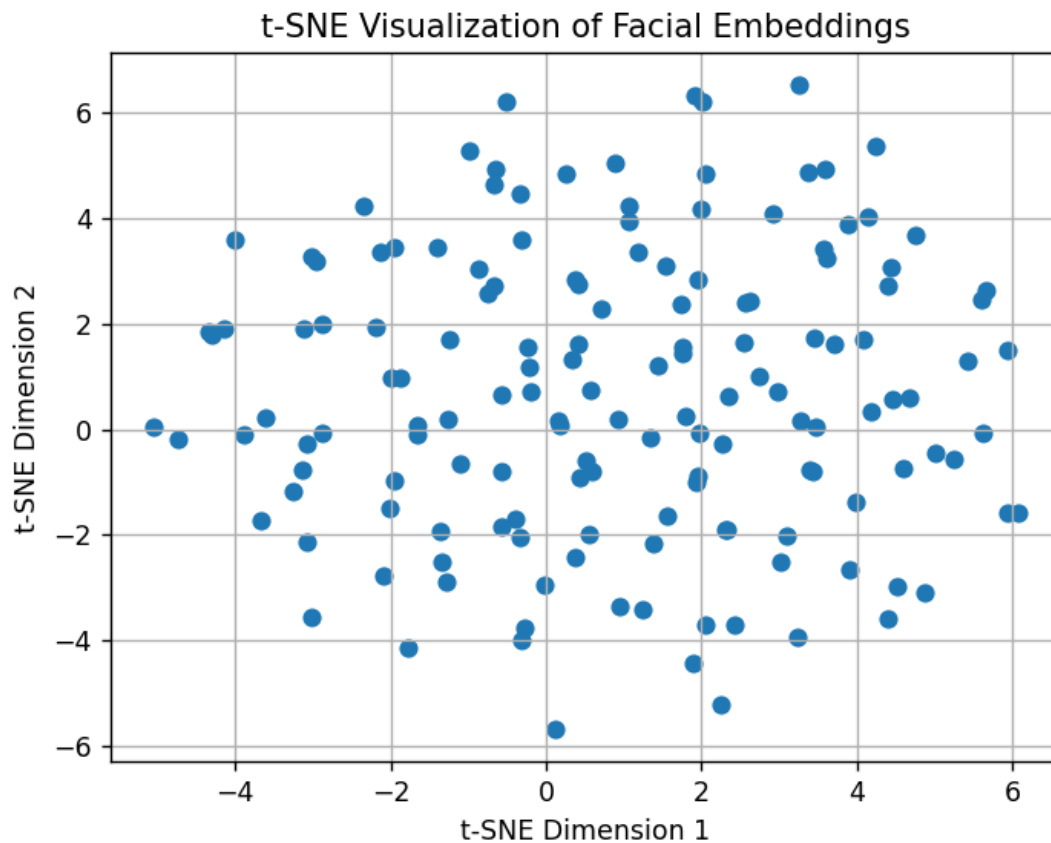
```
plt.show()
```

Why t-SNE is preferred over PCA for visualization:

PCA mainly preserves global linear variance, while t-SNE is designed to preserve local neighborhood relationships. Therefore, t-SNE can reveal complex clusters in high-dimensional data more effectively.

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12. Iris Dataset — t-SNE

Unit IV - Question 12

t-SNE on Iris Dataset

```
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_iris
from sklearn.manifold import TSNE
from sklearn.preprocessing import StandardScaler
```

```
iris = load_iris()
```

```
X = iris.data
```

```
y = iris.target
```

```
# Standardize
```

```
X_scaled = StandardScaler().fit_transform(X)
```

```
# t-SNE
```

```
tsne = TSNE(
    n_components=2,
    random_state=42,
    perplexity=30
```


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)

```
X_tsne = tsne.fit_transform(X_scaled)
```

```
# Plot
```

```
sns.scatterplot(  
    x=X_tsne[:, 0],  
    y=X_tsne[:, 1],  
    hue=iris.target_names[y],  
    palette="deep"  
)
```

```
plt.title("t-SNE Visualization of Iris Dataset")
```

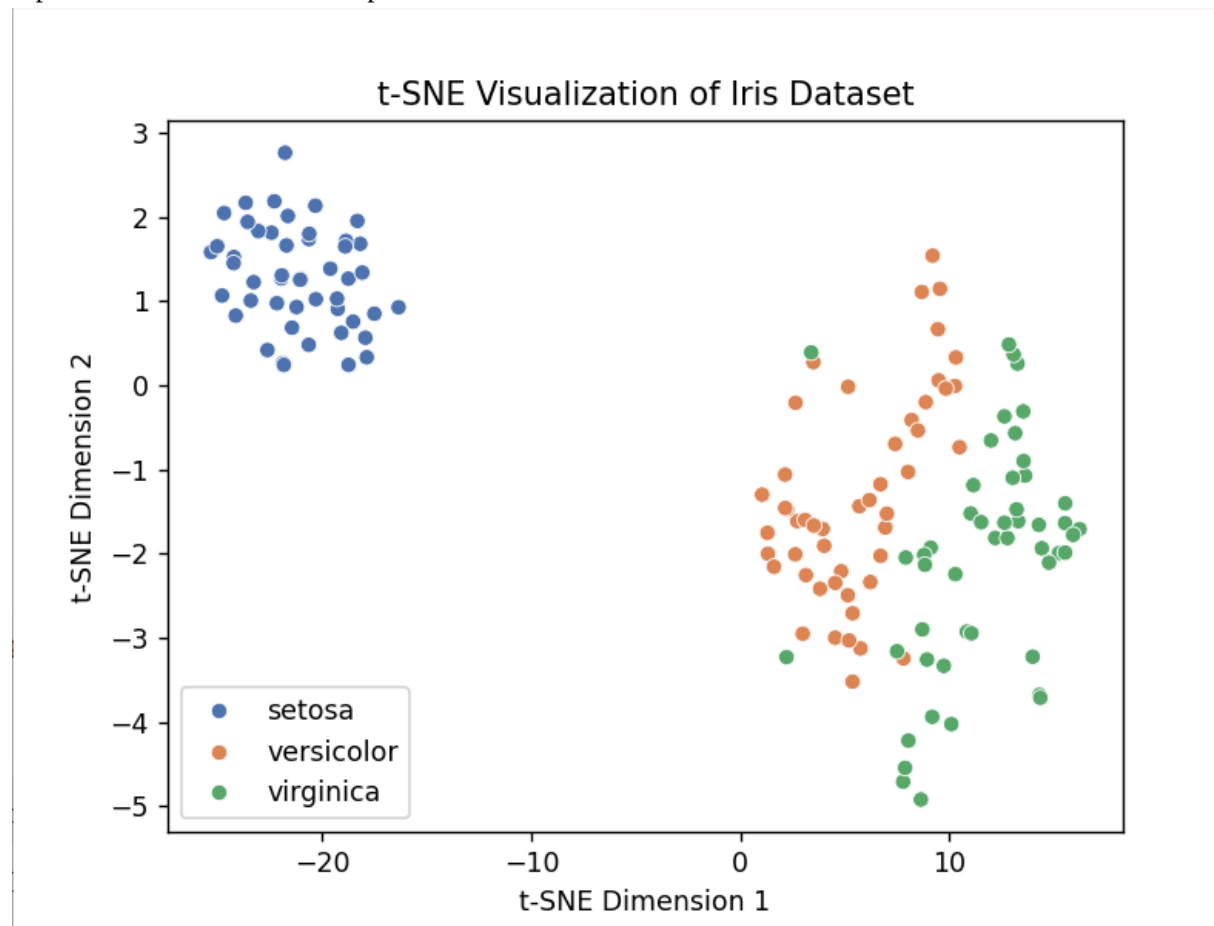
```
plt.xlabel("t-SNE Dimension 1")
```

```
plt.ylabel("t-SNE Dimension 2")
```

```
plt.show()
```

Interpretation:

The three Iris species form relatively distinct groups, especially Iris-setosa, which is usually well separated from the other two species.



13. Customer Segmentation — t-SNE

Unit IV - Question 13

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t-SNE on 40 Behavioral Variables

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.manifold import TSNE
from sklearn.preprocessing import StandardScaler
```

```
np.random.seed(42)
```

```
# 200 customers and 40 behavioral variables
data = np.random.rand(200, 40)
```

```
# Standardize
scaled_data = StandardScaler().fit_transform(data)
```

```
# t-SNE
tsne = TSNE(
    n_components=2,
    random_state=42,
    perplexity=30
)
```

```
data_tsne = tsne.fit_transform(scaled_data)
```

```
# Plot
plt.scatter(
    data_tsne[:, 0],
    data_tsne[:, 1]
)
```

```
plt.xlabel("t-SNE Dimension 1")
plt.ylabel("t-SNE Dimension 2")
plt.title("Customer Segmentation using t-SNE")
plt.grid(True)
```

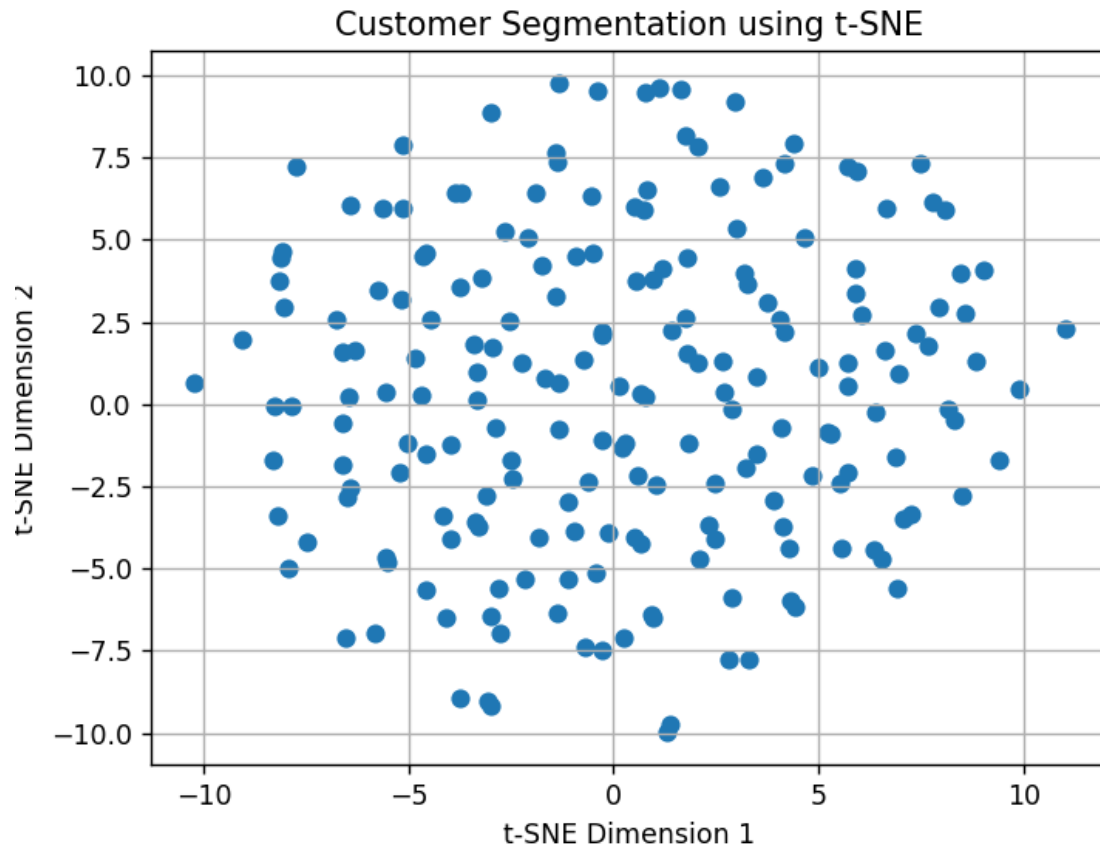
```
plt.show()
```

Discussion:

Distinct groups appearing as separated regions indicate potential customer clusters. If the points overlap heavily, the behavioral variables do not produce clearly separated groups.

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14. Medical Image Dataset — UMAP

First install UMAP if required:

```
pip install umap-learn
```

Then run:

```
# Unit IV - Question 14
# UMAP on Medical Image Features

import numpy as np
import matplotlib.pyplot as plt
import umap
from sklearn.preprocessing import StandardScaler

np.random.seed(42)

# 200 samples with 100 features
data = np.random.rand(200, 100)

# Standardization
scaled_data = StandardScaler().fit_transform(data)
```

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```
# UMAP
reducer = umap.UMAP(
    n_components=2,
    random_state=42
)

data_umap = reducer.fit_transform(scaled_data)

# Plot
plt.scatter(
    data_umap[:, 0],
    data_umap[:, 1]
)

plt.xlabel("UMAP Dimension 1")
plt.ylabel("UMAP Dimension 2")
plt.title("UMAP Visualization of Medical Image Features")
plt.grid(True)

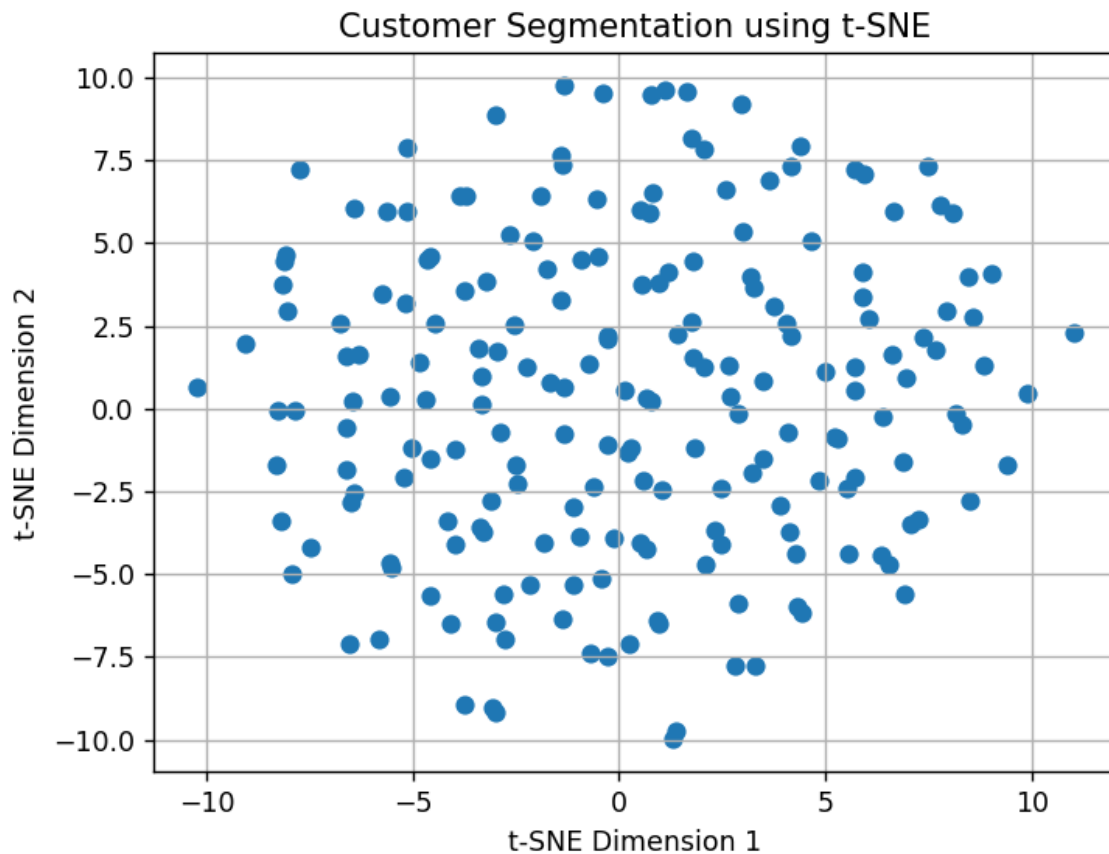
plt.show()
```

Comparison with PCA:

PCA is a linear dimensionality reduction technique, whereas UMAP can capture non-linear relationships and often provides better visualization of complex clusters.

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15. Iris Dataset — UMAP

Unit IV - Question 15

UMAP on Iris Dataset

```
import matplotlib.pyplot as plt
import seaborn as sns
import umap
```

```
from sklearn.datasets import load_iris
from sklearn.preprocessing import StandardScaler
```

```
iris = load_iris()
```

```
X = iris.data
y = iris.target
```

```
# Standardize
X_scaled = StandardScaler().fit_transform(X)
```

```
# UMAP
reducer = umap.UMAP(
    n_components=2,
    random_state=42
```

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)

```
X_umap = reducer.fit_transform(X_scaled)
```

```
# Plot
```

```
sns.scatterplot(  
    x=X_umap[:, 0],  
    y=X_umap[:, 1],  
    hue=iris.target_names[y],  
    palette="deep"  
)
```

```
plt.title("UMAP Visualization of Iris Dataset")
```

```
plt.xlabel("UMAP Dimension 1")
```

```
plt.ylabel("UMAP Dimension 2")
```

```
plt.show()
```

